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Promoting and Developing Sustainable Behavior among Engineering Students: The Role of Higher Education Institutions, Digital Environments and Digital Technologies

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This work contributes to the following Sustainable Development Goals (SDGs).



Sustainability competencies + Campus practices + digital innovations & Virtual worlds.

With the right combinations, a unified strategy can be a strong way to educate and influence towards better, more sustainable behavior.

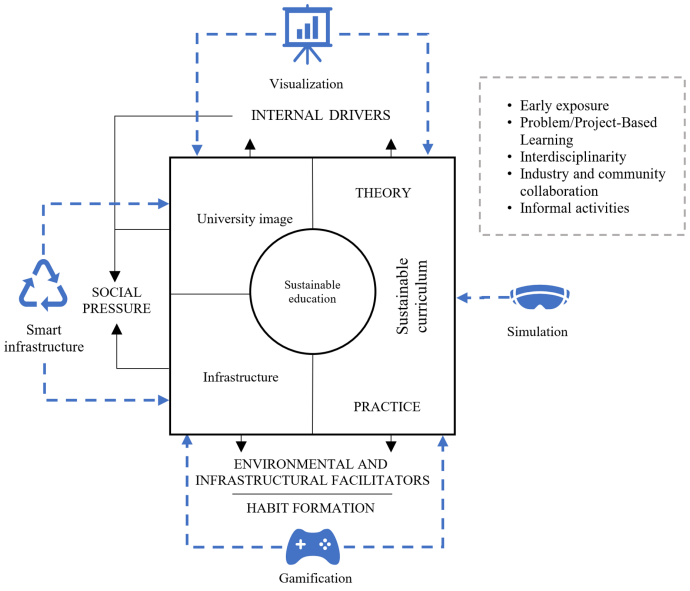
Open research gap:

Sustainability awareness does not translate into sustainable behavior

Universities can no longer view sustainability and digitalization as separate priorities.

We recommend:

- Integrating behavioral science models into sustainability education design
- Embedding digital technologies for sustainability education and students' well-being
- Conducting longitudinal studies to examine long-term behavioral change
- Exploring AI-supported systems, virtual reality, digital twins, and metaverse for sustainability, wellbeing, and soft skills training
- Investigating how social media environments influence students' sustainability attitudes, mental well-being, and behavior.



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Highlights of Sustainability

Review

Promoting and Developing Sustainable Behavior among Engineering Students: The Role of Higher Education Institutions, Digital Environments and Digital Technologies

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Abstract Our research explores how higher education institutions (HEI) can effectively foster sustainable behavior and better wellbeing among engineering students, using digital technologies to enhance development of these life aspects. Engineering graduates, as future professionals, influence technological development and contribute to workplace cultures, yet often lack the knowledge to address sustainability challenges, thereby increasing the risk of developing unsustainable solutions and work-life habits. Universities offer powerful learning, social, and living environments that can influence long-term attitudes and habits. To understand how these environments shape sustainable behavior and how they can be improved, a narrative literature review was conducted across behavioral science, sustainability studies, engineering education, and digital technology research. The review synthesizes four key categories of behavioral factors: internal drivers, social pressure, environmental and infrastructural facilitators, and habit formation. Universities can address these categories through their curricula, institutional image, and campus infrastructure. The analysis highlights several effective educational strategies, including early exposure to sustainability concepts, problem- and project-based learning, interdisciplinarity, collaboration with industry and communities, and informal activities that extend beyond the classroom. A second contribution of this study is the integration of technological tools into the sustainability education discussion. As digital technologies become deeply embedded in students' learning processes and increasingly shape how they access information and interact with systems and artificial intelligence tools to make decisions, Industry 4.0 and 5.0 tools serve as powerful mechanisms for promoting sustainable behavior. The review identifies four categories of digital tools that can support sustainable behavior development: visualization, simulation, gamification, and smart infrastructure. The study proposes an integrative conceptual framework that aligns pedagogical strategies, institutional image, and campus infrastructure to support the development of sustainable behavior among engineering students. We recommend follow-up studies on social media's impact on mental health, virtual and metaverse technologies for wellbeing and soft skills development, especially in an era of increasing reliance on artificial intelligence (AI) tools for human work and thinking offloading. Additionally, linking technological tools with behavioral predictors to demonstrate how technology can improve learning has been shown to boost social and motivational factors and incorporate sustainable actions into daily routines. This presents strong opportunities to connect HEI education to Society 5.0 and Education 5.0 development.

Keywords sustainability; behavior; HEI; higher education; engineering studies; psychological foundations; wellbeing; social pressure; accessibility; habit formation; curriculum design; digital technologies; Industry 5.0; Society 5.0; Education 5.0; digitalization

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1. Introduction

Higher education institutions (HEI) are uniquely capable of shaping students' development of sustainable behaviors because they offer the learning, socialization, and living environments for young professionals [1]. With positive surroundings, individuals can develop sustainable behaviors, which can also influence households. This, in turn, continues to positively affect surrounding communities and beyond [2]. This study explores how universities can encourage sustainable living, thinking, and action among engineering students by conducting a narrative literature review and offering actionable recommendations for future strategies, learning process

development, and technological integration for higher education institutions to develop their curricula, teaching methods, and staff guidance. The findings suggest that sustainable behavior is shaped by a range of factors, including internal drivers, social pressure, environmental and infrastructural facilitators, and habit formation. Universities can strategically build new brand images and address each of these layers through university image, sustainable curriculum design, and sustainable infrastructure [3]. The study provides a sustainable education framework linked to models of sustainable behavior formation. By implementing this framework, universities can guide students from awareness of sustainability to becoming responsible, sustainability-minded professionals who take action and know where to look for tangible positive impacts on the Sustainable Development Goals, with the proper knowledge and capacity.

The need to integrate sustainability into modern education is evident in the global context. While the importance of technology in society is growing, engineering professionals who support technological development often lack proper expertise in sustainability [4–7]. Consequently, the risk of creating unsustainable solutions increases, and an ever-growing historical load of unsustainable technology solutions accumulates. The lack of a sustainability focus in engineering education is linked to challenges in integrating sustainability practices into courses, including rigid curricula, limited faculty expertise, and a lack of institutional alignment [4,6–8].

Meanwhile, education plays a vital role in shaping sustainable attitudes and behaviors [9–11]. Students enrolled in sustainability-focused programs demonstrate greater environmental awareness and responsibility than those enrolled in traditional engineering curricula [12]. The role of universities extends beyond curriculum design. They can impact students not just through formal education but also through organizational culture, infrastructure, and hands-on experience. Recent studies show that this role can be fulfilled from multiple viewpoints, but this is not occurring systematically or equally across all sectors. A study by Masor et al. [13] shows a primary focus on environmental and social development (46%), followed by building and layout, administrative and governance, and quality of service (40%). However, there was limited literature on inclusivity (8%) and transportation and mobility (6%) in relation to sustainability. This shows that there are still extensive opportunities left on the table, where more could be done for a better shared future.

As sustainable education prepares engineers to focus on sustainability, empowering them to bring their values to the industry [5,10,14] and to society at large [15], this paper aims to demonstrate how universities can stimulate students' sustainable behavior and to explore how digital technologies can support and shape it. To achieve this goal, the study answers the following research questions:

RQ1: How does education influence students' sustainable behavior?
RQ2: What can universities do to support students' sustainable behavior?
RQ3: How can digital technologies and smart-campus solutions contribute to the development of sustainable behavior?

2. Related Studies

Sustainable behavior is a behavior targeted to achieve social good and includes aspects like environmental, social, technical, and economic sustainability [7,16,17]. The alternative terms used in the literature are altruistic behavior [18], pro-environmental [19], etc. It is worth noting that sustainability is frequently seen from an environmental perspective [10,20]. However, it is essential to recognize the importance of broadening the understanding of sustainability to include social, economic, and technical aspects, as often outlined in generalized sustainability classification models [21,22]. Several models explain how sustainable behavior develops, highlighting the significant influence of the surrounding culture and socio-economic environment on an individual's actions. The key ideas are summarized in Table 1.

Predicting behavior is challenging and yet not fully investigated [23]. For example, research indicates that the existing TPB model accounts for only 30% of pro-environmental behavior [24]. Furthermore, various behaviors are influenced by distinct factors. For instance, breaking down waste management into reuse, reduction, and recycling behaviors helps researchers pinpoint different influencing elements [25]. Interestingly, while the recycling experience has minimal direct impact on recycling behavior, it does increase the willingness to reduce and reuse waste [25]. Both awareness and experience shape attitudes and perceived behavioral control. This underscores the importance of addressing a wide range of behavioral predictors to improve outcomes

in pro-environmental initiatives. Consequently, this paper considers the significance of all factors and categorizes them into four groups for further analysis: internal drivers, social pressure, environmental and infrastructural facilitators, and habit formation. These categories are outlined in Figure 1.

Table 1. Sustainable behavior formation models.

The Theory of Planned Behavior (TPB)	The model suggests that behavior is influenced by attitudes, subjective norms, and perceived control, which together form intention and behavior [23,26]. Attitude reflects the favorability of the behavior; subjective norm refers to social pressure; perceived control reflects the ease of performance. Some scholars advocate including moral norms, habits, personal values, and situational constraints to capture the complexities of environmental decisions [25,27].
Value Belief Model (VBN)	The key elements of the model are individuals' *values* that form *beliefs* and then *personal norms* that lead to behavior [28]. This model does not focus on contextual barriers but on attitudinal factors.
Attitudes Behavior Context (ABC)	The expansion of the VBN model [28]. It acknowledges the role of context and indicates that the difficulties of performing the behavior—such as time, costs, efforts, and other aspects—diminish the impact of attitudinal factors.
Norm Activation Theory (NAM)	The crucial factors include awareness, which influences responsibility and shapes personal norms, along with guilt that drives behavior. This model focuses on individual morality and excludes contextual elements [18,19].
Comprehensive Action Determination Model (CADM)	A combination of the models [29] that, on the first level, has attitudes, personal norms, perceived behavior control, and social norms, which form the intentions together. On the second level, one of the strongest predictors of behavior control based on this model is habit strength, which emphasizes the importance of prior experience for behavior adaptation [29].

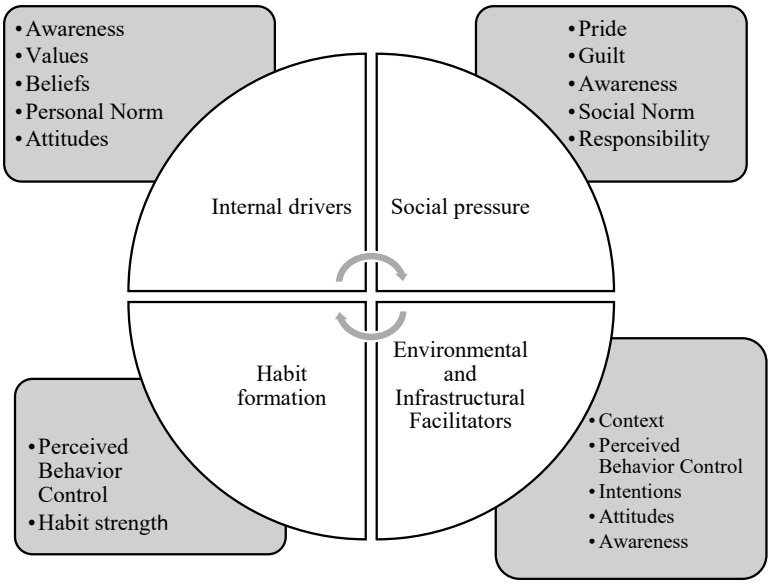


Figure 1. Factors of sustainable behavior.

2.1. Internal Drivers

Internal drivers represent the strength of sustainable values. Studies indicate that students who hold positive views on sustainability and regard it as ethically significant are more inclined to engage in sustainable practices [4,12]. Those who prioritize nature and socially responsible values tend to adopt sustainable behaviors as well [5,14]. Educational initiatives that emphasize the impact of unsustainable practices can stimulate personal values and promote sustainable actions [4,5,12].

2.2. Social Pressure

Social pressure represents the social environment and the effects of individuals' sense of duty on behavior. Social influence plays a role in behavior formation: students engage more in sustainable actions when peers and institutions support them [5,6] or there is a strong social norm supporting sustainability [30].

2.3. Environmental and Infrastructural Facilitators

Environmental and infrastructural facilitators are the likelihood of performing the behavior based on the availability of resources, perceived control of the situation, and the desire to take action. Although sustainability education helps develop strong intentions, actual behavior depends on the environment [7]. Individuals may hold positive views on sustainability, yet they may not act sustainably if the context is unsupportive (such as insufficient recycling infrastructure or cost barriers). Perceived behavioral control affects actual participation; for instance, students tend to recycle more when recycling bins are easily accessible [30]. Behavioral nudging (e.g., visible recycling bins and reminders) can help align behavior with attitudes [31,32].

2.4. Habit Formation

Habit formation is the likelihood of performing a behavior based on prior experience. While raising awareness and fostering positive attitudes toward sustainability are essential first steps, they are often insufficient alone to generate sustained pro-environmental behavior. Practical experience plays a key role in "activating" or bridging the gap between attitude and behavior [10,11,24,30,33]. Students who established early sustainable consumption habits, such as using reusable items, tended to sustain these practices in their careers [12].

3. Method

To investigate how higher education institutions influence engineering students' sustainable behavior and how digital technologies can support this process, a narrative literature review was conducted. A narrative approach was selected because it allows the integration of findings from multiple disciplines: behavioral science, sustainability studies, engineering education, and digital technologies. A notable aspect of this method is its selectivity, as researchers select studies based on their relevance rather than sticking to systematic search criteria. This enables mapping the landscape of a complex topic [34,35].

The primary databases used were Scopus, Web of Science, and Google Scholar, as they provide broad coverage of peer-reviewed journals and conference proceedings in engineering, education, and sustainability research. Additional relevant publications were identified through backward snowballing, which implies using the reference lists of key articles [36].

The search focused on studies published between 2010 and 2024, a period that captures the growing integration of sustainability into engineering education and the emergence of technological tools in educational and campus contexts. This timeframe was chosen because the year 2010 marked a tipping point for increasing interest in research on circularity and sustainability, which has been growing exponentially since then. Foundational theoretical work on environmental and altruistic behavior (e.g., Theory of Planned Behavior, Norm Activation, Value-Belief-Norm models) published before 2000 was also included due to its conceptual importance. The selected period also reflects major global policy developments, particularly the adoption of the United Nations Sustainable Development Goals (SDGs) in 2015 [37], which significantly accelerated research and institutional initiatives related to sustainability in higher education.

3.1. Search Strategy

A broad but structured search strategy was adopted. Combinations of keywords were used to capture both behavioral and educational/technological aspects, including but not limited to:

- "sustainable behavior", "social responsibility", "pro-environmental behavior", "environmentally significant behavior", "recycling behavior", "waste management";
- "engineering education", "STEM education", "sustainability in engineering curriculum";
- "IoT", "smart campus", "smart building", "digital tools", "gamification", "technology".

Studies were selected based on their relevance and contribution to conceptual clarity. The focus was on the diversity of viewpoints and the depth of analysis, consistent with the narrative

review methodology. Studies were included if they met the following conditions:

1. Focus on sustainable behavior or behavior-related constructs: the study examined predictors or outcomes of sustainable/pro-environmental or socially responsible behavior.
2. Relevance to engineering higher education: the study was conducted in a higher education setting and focused on engineering/STEM students, curricula, or educational interventions related to sustainability.

To enhance the framework, additional studies on digital tools and smart systems in educational or campus settings related to sustainability, energy, or environmental awareness were reviewed. Digital technologies are essential to consider since they are now deeply integrated into higher education, shaping the learning experience, and can also promote sustainable behaviors by highlighting environmental impacts, enabling data-driven analysis, and supporting eco-friendly campus infrastructure. Examining their role offers a more accurate, future-oriented view of sustainability education.

The search followed an iterative narrative review process rather than a strictly systematic protocol. Initial searches were conducted using combinations of predefined keywords, and records were screened against the inclusion criteria based on their titles and abstracts. Full-text articles of potentially relevant studies were then examined in detail. The search strategy was refined iteratively as new concepts and relevant terms emerged from the literature. This approach aligns with the narrative review methodology, which prioritizes conceptual coverage and diverse perspectives over exhaustive retrieval [34,35], offering valuable insights into the status quo and serving as a starting point for topic- and context-specific, focused systematic literature reviews.

3.2. Synthesis of Findings

The analysis followed a theory-informed thematic synthesis [38]. First, key behavioral frameworks identified in the literature (including the Theory of Planned Behavior, Value-Belief-Norm model, Norm Activation Model, and Comprehensive Action Determination Model) were used to identify core behavioral determinants. Grounded in the theories, these determinants were grouped into four categories: internal drivers, social pressure, environmental and infrastructural facilitators, and habit formation.

In the second stage, studies on sustainability education practices and digital technologies were analyzed and mapped to these behavioral categories to link potential interventions with theories of behavior change. This alignment enabled the integration of educational and technological interventions with underlying behavioral predictors and informed the development of the conceptual framework presented in Figure 2, which explains the main avenues of tertiary education development to enable change in students' sustainable behavior.

When studies reported contrasting findings, these differences were explicitly documented and analyzed as indicators of unresolved tensions in the literature. Rather than resolving inconsistencies through exclusion, they were used to identify knowledge gaps and inform the critical discussion and future research directions.

4. Literature Review

Based on these findings, HEIs have three potential directions for shaping students' sustainable behavior: providing sustainable curricula (with both theoretical and practical learning), setting an example of being a sustainable organization, and providing sustainable infrastructure to ease sustainable behavior. The connection between sustainable education action directions and sustainable behavior factors is presented in Figure 2.

4.1. Sustainable Curriculum

Although education can increase environmental awareness, attitudes, and, subsequently, behavior [10,12,27], many students report that sustainability topics are either mentioned superficially or overlooked in their courses [5,6,14,39]. Certain fields integrate sustainability concepts more naturally, while others struggle because of curriculum restrictions, a lack of faculty expertise, or disciplinary traditions [4–7]. For example, Software Engineering programs rarely address sustainability [7]. Often, sustainability learning is provided as an extracurricular activity with self-study materials, online learning, and independent research [40].

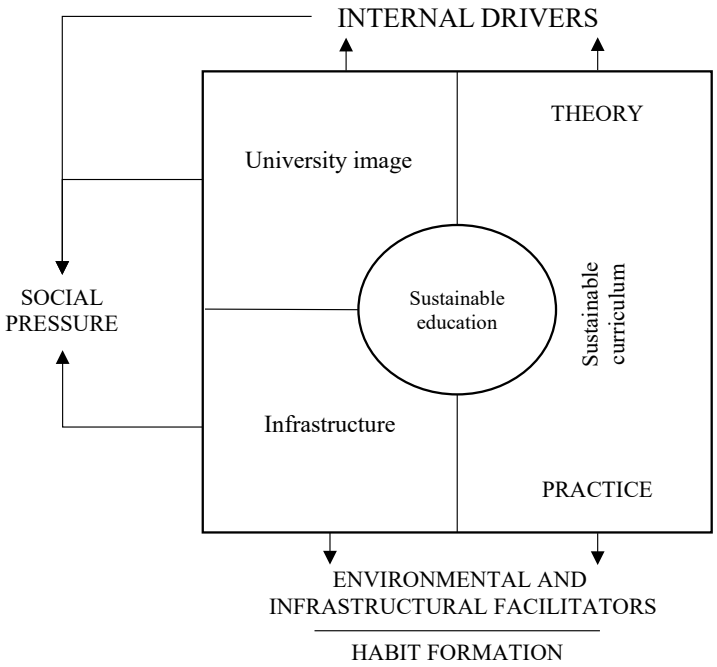


Figure 2. Sustainable education framework.

These results highlight an urgent need for a framework that integrates sustainable development into the curriculum. Based on the literature, integrating technologies can enhance sustainability education across various application areas [41–43]. This section offers guidance for universities and academics on adapting sustainability principles within engineering fields and provides examples of technological tools that can support sustainability learning.

Based on the literature, several key directions for sustainable curriculum development were identified. These strategies are proven to enhance students' understanding, values, and engagement with sustainability across engineering fields. They involve early exposure to sustainability concepts, problem- and project-based learning, interdisciplinarity, collaboration with industry and communities, and informal learning activities. Together, these elements create a pathway that fosters both conceptual understanding and practical experience, helping universities to integrate sustainability more systematically and effectively into engineering education.

4.1.1. Early Exposure

There is an ongoing debate about introducing sustainability to students at the start of their education [6] versus after they have acquired foundational knowledge from the traditional curriculum [4,7]. Engineering educators worry that first-year students may find sustainability concepts too abstract and struggle to understand their relevance without practical engineering experience [7].

Nonetheless, providing early exposure to sustainability concepts can highlight the social importance of abstract subjects, such as mathematics, and demonstrate their potential to address real-world issues. This approach can integrate sustainability into students' professional mindset rather than treating it as an additional topic later on [6,12]. Research indicates that students who engage with sustainability concepts early develop a stronger commitment to pro-environmental behavior and responsible consumption [12], recognizing it as an essential engineering competency [10]. Furthermore, sustained involvement in sustainability initiatives increases the likelihood that students will engage in pro-environmental actions after graduation [9,32].

4.1.2. Problem-/Project-Based Learning

Studies highlight that formal education significantly enhances students' understanding of sustainability issues, but awareness alone does not always translate into action [9,12,39]. Many students feel a lack of real-world applicability in sustainability courses [6,27]. Therefore, it is recommended that problem-/project-based learning be introduced in the sustainable curriculum [40] as active student engagement in sustainability-focused projects expands their perception of the engineering role from industrial production to social responsibility [44].

Engineering students tend to prioritize economic and technical factors over sustainability [4,15]. Even if they consider environmental impacts in design, ethics and social responsibility are still frequently overlooked [14,40]. With a proper design, sustainability-focused engineering projects develop a deeper appreciation for environmental, social, and economic factors [6,45]. Working on real-world projects helps students to reflect on their values and ethical considerations in decision-making [5,14] in addition to getting hands-on experience in sustainability challenges [20,40].

4.1.3. Interdisciplinarity

Interdisciplinarity is essential for effective sustainability education, as sustainable solutions require knowledge from multiple fields, including engineering, social sciences, environmental science, economics, and policy-making [5,6,14,45]. Moreover, interdisciplinary exposure helps students develop systems thinking and ethical decision-making skills [40].

To give students interdisciplinary experience, researchers recommend cross-disciplinary case study learning [6], collaborative learning between engineering, business and social sciences students [4], cross-disciplinary curriculum integrations (e.g., engineering ethics course incorporating discussions on social justice and environmental policy) [5] and joint/dual degree programs (a mechanical engineering student minoring in sustainability studies to understand life-cycle assessments) [20].

While engineering expertise is crucial for designing sensing architectures and AI-driven algorithms [46–48], effective implementation relies on insight from economics, psychology, and public policy to ensure real-world viability and user adoption [44,49].

This interdisciplinary approach not only improves problem-solving outcomes but also enriches student learning by providing practical exposure to how distinct fields converge to address the environmental, social, and technological dimensions of sustainability.

4.1.4. Industry and Community Collaboration

Working together with higher education institutions, industry, and local communities is vital for cultivating sustainability skills in engineering students. Experiential learning, where students engage in real-world sustainability issues alongside communities or organizations, has proven to enhance sustainability awareness and social responsibility [20,40]. Engineering graduates with industry experience in sustainability initiatives demonstrate higher environmental awareness and problem-solving skills [12].

Some universities integrate sustainability into fieldwork and internships, providing direct exposure to environmental issues [45]. Companies that offer sustainability internships see increased student interest in environmental innovation [7,50]. They benefit by gaining fresh insights from students while universities get access to real-world case studies [4,7].

Community engagement fosters ethical awareness and the ability to work in interdisciplinary teams [5,51]. Collaborative efforts between students and local governments drive community participation in environmental initiatives [24,45] and help to create sustainable urban planning solutions that directly benefit local communities [6]. The inclusion of external stakeholders in teaching and curriculum design also supports sustainability as a shared societal responsibility, reinforcing the interdisciplinary and collective nature of the issue [7,48].

4.1.5. Informal Activities

Studies show that extracurricular activities such as sustainability forums, competitions, campus audits, and volunteering in environmental initiatives help students develop practical skills and habits related to sustainability [7,20,39,52], leading to increased sustainability awareness levels compared to final-year engineering students [12]. Therefore, additional university activities may boost students' adaptations to sustainable practices.

One example of such an activity could be a hackathon focused on sustainability. As mentioned earlier, Software Engineering education rarely addresses sustainability [7]. Meanwhile, hackathons are time-limited, project-based computing competitions. There are numerous examples of hackathons that have a sustainability context [53,54]. Due to their intensive nature, these events can help introduce students to solving socially significant problems with less effort from educators compared to designing a semester- or year-long course.

4.2. University Image and Infrastructure

Universities, and our school system in general, can serve as role models for students and employees by implementing sustainable practices in their operations and engaging in sustainability-related projects and collaborations [11,24,45,55]. These actions help to build sustainable mindsets [5] and a desire to be associated with the institution as a leader in sustainability [30,56].

Students often learn sustainability through campus-wide initiatives like waste reduction programs, energy-saving campaigns, carbon footprint reduction, and sustainable infrastructure [20]. Incorporating sustainability principles in campus design, such as green buildings, urban gardens, and renewable energy sources, demonstrates institutional commitment and encourages students to adopt sustainable practices [20,45].

Proper campus development increases the likelihood of sustainable habit formation. Despite high awareness of sustainability, students may fail to act sustainably due to inadequate infrastructure (e.g., insufficient or inconvenient recycling bins) [30]. Recycling facilities, water conservation systems, and energy-efficient buildings supported by smart tools provide an efficient, sustainable environment [30,31].

The following principles are recommended when planning sustainable infrastructure:

- *Clear Signage and Visual Cues.* Studies have shown that well-designed signs and labels significantly improve recycling behaviors by reducing confusion over waste sorting [30–32]. That could be color-coded bins and informational posters near waste disposal areas [30]. Behavioral psychology suggests that placing sustainability-related messages in high-traffic areas, such as dormitories, cafeterias, and classrooms, reinforces sustainable habits [31].
- *Gamification.* Competition elements and rewards can enhance sustainable activities engagement [31,56]. For example, universities can offer discounts on reusable coffee cups and sustainable purchases reward points. Collaborative and goal-driven competitions, e.g., inter-dormitory recycling challenges or footprint reduction competitions, encourage students to adopt long-term waste reduction habits [30–32].
- *Digital Tools and Smart Technologies.* IoT and AI-based systems that adjust lighting, heating, and cooling based on occupancy and environmental conditions significantly reduce energy waste by monitoring consumption [31,57–59]. In addition, mobile apps and interactive platforms help students monitor and improve their sustainability actions [20,31,32]. Sensor-equipped bins notify facilities when they are full, preventing overflow and optimizing waste collection [30,60].
- *Placement and Accessibility.* The location and accessibility of sustainable options should be seamless. For example, strategically placing recycling bins in visible and convenient locations (e.g., near exits and cafeterias) increases recycling rates [30]. Default sustainability features, such as automatic double-sided printing, promote eco-friendly practices seamlessly [31].

While most of the recommendations above focus on environmental sustainability, other aspects of sustainability should also be considered in infrastructure. Universities should strive to provide a safe and supportive environment for all students and to consider multiple dimensions of sustainability and the Sustainable Development Goals, with a focus on interactions [61]. For example, in terms of gender equality, placing a daycare in a university building can help students with children continue their education without interruption [62,63].

4.3. Digital Tools Integration

Building on the sustainable education framework, this study examines the role of technologies by showing how various tools can support the development of students' sustainable behavior. Based on the literature, four main categories of technological tools were identified (visualization, smart infrastructure, simulation, and gamification), each affecting different parts of the framework. Each category supports specific behavioral predictors by influencing internal drivers, perceived ease of action (environmental and infrastructural facilitators), habit formation, and motivational social pressure.

4.3.1. Visualization

Visualization tools such as dashboards, mobile apps, and real-time displays help students understand environmental impacts and see the outcomes of their actions. These tools primarily

target internal drivers by increasing awareness and strengthening sustainability-related attitudes. At the same time, access to real-time information enhances perceived behavioral control by helping users understand how their actions influence the system. IoT-enabled smart campus projects let students collect and analyze real-time data to optimize waste and energy management [46,57–59,64]. Public dashboards, such as energy-use visuals, comparison charts, or sustainability indicators, enhance transparency and boost the university's sustainability image. Studies on energy monitoring and feedback systems show that real-time data access increases engagement and supports data-driven decision-making [65–67].

4.3.2. Gamification

Gamification introduces reward structures, competition, and collaborative challenges that motivate both formal and informal sustainable behavior. This category shapes social pressure and habit formation. Leaderboards and public performance comparisons establish social norms and peer influence, encouraging students to align their behavior with group expectations. Repeated challenges and rewards sustain engagement and reinforce daily actions, supporting long-term habit development. Many universities use campus-wide digital competitions, such as energy-saving contests and recycling challenges, to increase engagement and foster long-term habits [65–67].

4.3.3. Simulation

Simulation technologies, such as digital twins and interactive modeling tools, allow students to explore sustainability challenges in safe, controlled, and highly interactive environments [68]. Simulation targets internal drivers and perceived behavioral control, supporting experiential learning and systems thinking. By allowing students to test design decisions and observe long-term environmental and social consequences, these tools strengthen understanding, values, and confidence in applying sustainable solutions. Simulation-based learning environments improve decision-making skills and deepen understanding of complex sustainability trade-offs [69].

4.3.4. Smart Infrastructure

Smart infrastructure integrates IoT, AI, and sensing technologies within the physical campus environment to improve operational efficiency [35]. This category mainly addresses environmental and infrastructural facilitators and habit formation by embedding sustainable behavior into everyday contexts and reducing the effort required to act sustainably. Systems with sensors, such as smart HVAC monitoring, occupancy-based lighting, air quality sensors, and smart waste bins, enable real-time resource management and make sustainable choices more convenient [14,20]. Evidence from smart campus implementations shows that automated feedback and convenient infrastructure increase participation and normalize sustainable routines over time [27,57]. At the same time, visible digital infrastructure reinforces the university's sustainability image and provides authentic data for use in coursework, research, and community collaborations.

Figure 3 illustrates how these categories of digital tools interact with the sustainable education framework to support the development of sustainable behavior among engineering students. Visualization strengthens internal drivers by increasing awareness and understanding; simulation supports internal drivers and perceived behavioral control through experiential learning, and allows safe environments for multiple repetitions; gamification reinforces social pressure, feeling of achievements and habit formation through competition and repeated engagement with use of "reward functions"; and smart infrastructure enhances environmental and infrastructural facilitators by embedding sustainable actions into everyday routines. Through these mechanisms, digital technologies reinforce the university's sustainability image, facilitate the integration of sustainable actions into students' daily academic and campus experiences, and provide motivating, engaging environments for learning about sustainability from multiple perspectives.

Visualization technologies help to make environmental impacts visible and understandable. Through dashboards, real-time data, and performance indicators, visualization tools reinforce the theoretical dimension of the sustainable curriculum and strengthen the university's sustainability image. Simulation tools support formal learning. Digital twins, modeling tools, and virtual experimentation environments allow students to engage with sustainability challenges in controlled, interactive settings, thereby deepening theoretical and practical knowledge. Gamification

motivates engagement through competition, rewards, and collaborative challenges. Gamified activities facilitate both formal and informal learning, strengthening habit formation and fostering long-term sustainable behavior. Smart infrastructure increases visibility of pro-environmental actions and, therefore, supports the university's image. At the same time, it facilitates integrating sustainable actions within daily routines.

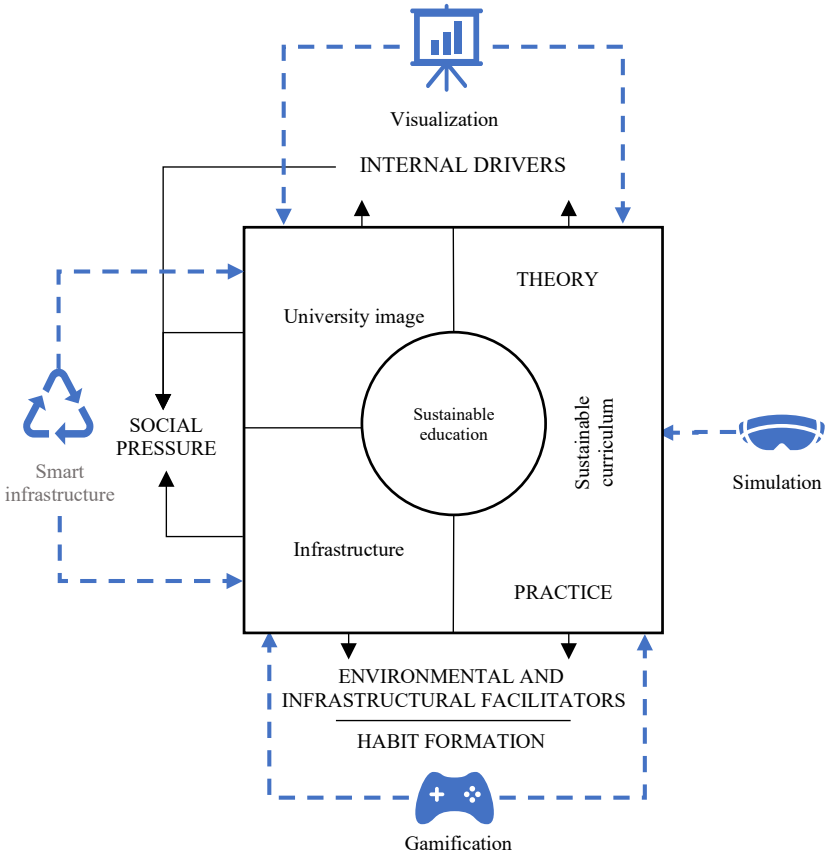


Figure 3. Integration of technologies into the sustainable education framework.

5. Discussion

This study investigates the nature of sustainable behavior and how higher education institutions can cultivate such behavior among engineering students. A review of behavioral formation models reveals that sustainable behavior is influenced by multiple interrelated factors, including internal drivers, social pressure, environmental and infrastructural facilitators, and habit formation. This study addressed three research questions concerning the role of higher education and digital technologies in supporting students' sustainable behavior.

RQ1: How does education influence students' sustainable behavior?

The findings indicate that education influences sustainable behavior through multiple pathways. Formal learning activities strengthen internal drivers by increasing awareness, knowledge, and sustainability-related values, whereas experiential approaches, such as project-based learning, support habit formation and perceived behavioral control through practical experience. Whether intentions translate into behavior depends on environmental and infrastructural facilitators: when campuses provide visible and convenient sustainable options, students face fewer practical barriers and are more likely to act on what they learn. Moreover, institutional culture and visible sustainability initiatives contribute to social norms, reinforcing sustainability as an expected professional and community standard.

RQ2: What can universities do to support students' sustainable behavior?

Therefore, universities can influence behavior through three complementary avenues: integrating sustainability systematically into the curriculum through early exposure, experiential learning, and interdisciplinary approaches; building a strong pro-sustainability institutional

image that reinforces social expectations; and providing supportive campus infrastructure that makes sustainable choices convenient and accessible.

RQ3: How can digital technologies and smart-campus solutions contribute to the development of sustainable behavior?

A growing body of research emphasizes the importance of integrating digital technologies into university-level learning activities. For example, IoT-based projects allow students to track energy consumption on campus and evaluate the impact of behavioral changes, fostering data-driven decision-making. Gamified platforms, as seen in studies like GAIA [66], have been shown to increase student engagement and promote energy-saving behaviors through competition and social learning. Digital technologies support sustainable behavior by strengthening specific behavioral mechanisms. Visualization tools increase awareness and support internal drivers; simulation enhances understanding and perceived behavioral control; gamification activates social norms and habit formation; and smart infrastructure embeds sustainable choices into everyday practices by improving environmental and infrastructural facilitators. In this way, digital technologies extend sustainability education beyond the classroom and integrate learning with daily experiences. Learning about digital tools and technologies' contributions to sustainability and circularity will help students understand the possibilities of being both economically and socially competitive [70,71].

While the identified pedagogical strategies are widely recognized as effective, their large-scale implementation presents some acknowledged potential challenges too, which include: limited base curriculum space, insufficient faculty expertise on different aspects of sustainable development, the complexity of managing real-world projects alignment with educational progress requirements, and difficulties in coordinating interdisciplinary collaboration.

Still, digital technologies can also help to address these challenges. Online collaboration platforms and project management tools can support scalable implementation of problem- and project-based learning by facilitating teamwork, monitoring progress, and enabling remote interaction with industry or community partners [72]. Simulation environments and digital twins allow students to engage with complex sustainability scenarios without the logistical constraints of physical projects [73]. Visualization dashboards provide shared data resources that can be used across multiple courses, reducing preparation effort while maintaining real-world relevance. Digital Avatars, which are connected to a large language model (LLM) engine with sustainability-trained contextual knowledge, enable teachers and universities to provide sustainability support to their students and to serve as digital teaching assistants in courses. By reducing organizational complexity and increasing scalability, digital tools help translate sustainability pedagogy from isolated initiatives into institution-wide practice.

Beyond their individual effects, the four behavioral domains identified in this study (internal drivers, social pressure, environmental and infrastructural facilitators, and habit formation) should not be viewed as independent influences but as interacting elements within a dynamic system. Strong internal drivers, such as personal values or moral norms, may motivate students to act sustainably even when environmental facilitators are limited. Likewise, supportive environments and visible institutional practices can compensate for initially weak motivation by reducing effort and normalizing sustainable actions, thereby gradually strengthening internal attitudes and habits. Social pressure also interacts with individual and contextual factors: peer norms and institutional expectations can reinforce personal commitment and encourage behavior even when individual motivation is moderate. Understanding these interactions underscores the importance of coordinated interventions that simultaneously address individuals' motivation, social context, and environmental conditions.

6. Conclusion

This narrative review aimed to understand how higher education institutions can promote sustainable behavior among engineering students and how digital technologies can support this process. Across the literature on pro-environmental and altruistic behavior, four interconnected groups of predictors emerged: internal drivers (values, attitudes, moral norms), social pressure, environmental and infrastructural facilitators, and habit formation. The review reveals that universities are uniquely positioned to address all four, not only through what they teach but also through how they operate and the environments they create for students.

The findings highlight three primary paths for institutional action: sustainable curriculum

development, fostering a pro-sustainable university image, and creating supportive infrastructure. Among the curriculum strategies, five key approaches stand out as especially effective: early exposure to sustainability concepts, problem and project-based learning, interdisciplinarity, collaboration with industry and communities, and informal activities such as hackathons, volunteering, or student-led initiatives. These methods link theoretical knowledge with practical experience and help students transition from awareness to becoming committed, sustainability-focused professionals.

A second important contribution of this work is the integration of digital tools into the educational landscape. In the modern societies, where digital technologies increasingly shape how students access information and interact with systems and artificial intelligence tools to make decisions, Industry 4.0 and 5.0 tools serve as powerful mechanisms for promoting sustainable behavior. The review identifies four categories of technologies that correspond to different aspects of the sustainability education framework: visualization, simulation, gamification, and smart infrastructure. Visualization tools make environmental impacts visible and support both the university's image and learning experience. Simulation enhances the curriculum by enabling students to experiment with complex socio-technical systems in safe environments. Gamification supports both formal and informal learning by reinforcing social norms and fostering habit formation through motivation, feedback, and competition. Smart infrastructure incorporates sustainability into daily campus life, boosting perceived behavioral control and making sustainable choices easier.

Figure 4 integrates the sustainable education framework with curriculum strategies and digital technologies, illustrating how different institutional actions influence the development of sustainable behavior. At the center of the diagram, sustainable education represents the core outcome that emerges from the alignment of three institutional components: sustainable curriculum, university image/brand, and campus infrastructure

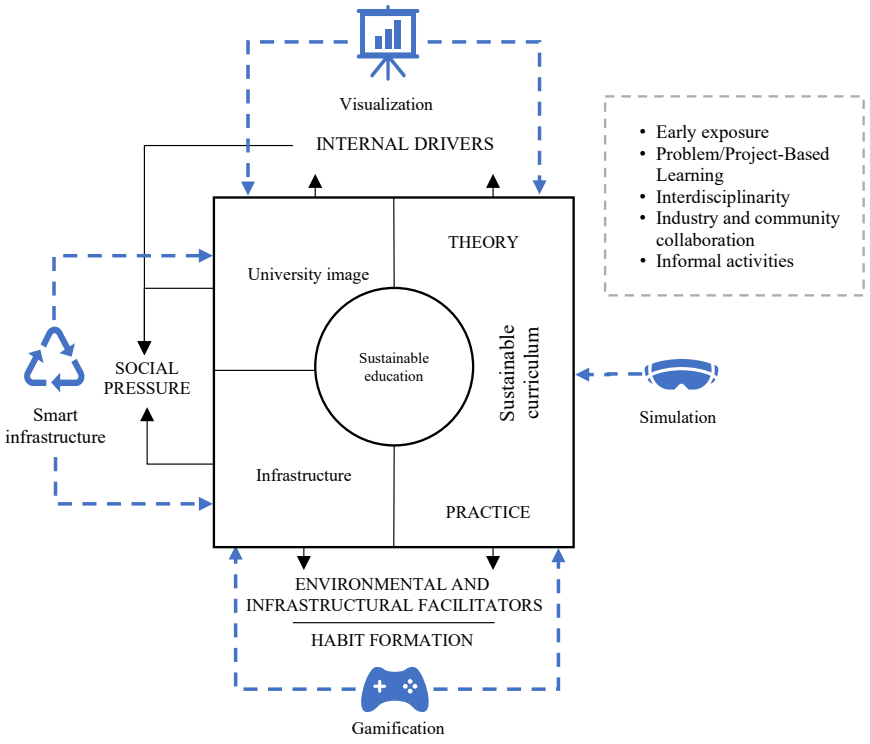


Figure 4. Sustainable behavior development in higher education.

Around this core, the four identified behavioral domains (internal drivers, social pressure, environmental and infrastructural facilitators, and habit formation) are positioned to indicate how educational interventions influence student behavior. Arrows between the institutional components and these domains illustrate how curriculum design, institutional culture, and infrastructure contribute to the formation of sustainability-related attitudes, norms, perceived control, and daily practices.

The dashed areas surrounding the framework represent the role of digital technologies. Each

technology category is positioned aligning to the behavioral domain it primarily supports: visualization strengthens internal drivers by increasing awareness and understanding; simulation supports the curriculum by enabling experiential learning and improving perceived behavioral control; smart infrastructure enhances environmental and infrastructural facilitators and reinforces social norms through visible institutional practices; and gamification promotes social influence and habit formation through competition, feedback, and repeated engagement.

The boxed list on the right specifies the key pedagogical strategies recommended for the sustainable curriculum, including early exposure, problem/project-based learning, interdisciplinarity, industry and community collaboration, and informal activities. These elements demonstrate how curriculum, institutional practices, and technological systems interact to enable the development of sustainable behavior. They illustrate how well-integrated, well-planned systems can be used to build more motivating and inspiring learning environments, while the university simultaneously contributes to knowledge growth and sustainable activity among its students and staff.

Overall, the evidence indicates that universities can no longer view sustainability and digitalization as separate priorities. Instead, combining sustainability competencies, campus practices, and digital innovation into a unified strategy provides a strong way to influence the behaviors of future engineers, specially right now when we live through extensive artificial intelligence integration into our societies [74–76] and at the same time higher educational units are seeking ways to boost the positives of it in education and mitigate the problems too [77].

This study has several limitations that also identify important directions for future research. As a narrative literature review, the analysis involves selective interpretation and may reflect researchers' judgments in identifying and synthesizing sources. In addition, the reviewed studies represent diverse cultural and institutional contexts in which the relationships between behavioral predictors and educational interventions may vary across settings [2].

The review reveals several tensions and gaps in existing research. While some studies report that early exposure strengthens long-term commitment to sustainability [12], others suggest that students in the early stages of engineering education may lack sufficient disciplinary context to engage meaningfully with sustainability concepts [6,7].

To progress the field, future studies should focus on closing the known gap between sustainability awareness and actual behavior. Even well-established models, such as the Theory of Planned Behavior, explain only a limited portion of behavioral variability [30], underscoring the need for longitudinal, context-sensitive evaluation of educational interventions. Additionally, with sustainability awareness and actual behavior link, academia needs to develop new pedagogical innovations [78], and practical strategies to address ways to operationalize sustainability education as part of study programs.

Research on digital technologies in sustainability education is largely technology-centered. Existing studies do not systematically link interventions to behavioral theory or comprehensive educational frameworks [57,59,65,67]. The framework proposed in this study addresses this conceptual gap, but its practical effectiveness has not yet been empirically validated.

Therefore, systematic empirical research is encouraged in the future to open new avenues for practical approaches to integrating different aspects of sustainability into the daily operations of higher education institutions. For example, longitudinal case studies could be used to track changes in students' attitudes, intentions, and behaviors during the implementation of integrated initiatives, such as smart campus systems or data-driven sustainability programs. Such studies would help determine whether educational and technological interventions lead to sustained behavioral change rather than short-term awareness.

Second, quasi-experimental studies could compare student outcomes in courses that incorporate the proposed digital tool integrations, such as visualization dashboards, simulations, or gamified challenges, with those delivered through traditional approaches.

Third, design-based research could support the iterative development and refinement of specific interventions, allowing adaptation across disciplines and institutional contexts.

An additional question that remains insufficiently addressed in the literature concerns whether different dimensions of sustainability (social, economic, technological, and environmental) require distinct educational and technological approaches. Most existing studies focus on environmental outcomes, such as energy consumption or waste reduction, while social and economic sustainability receive comparatively less attention [21,22]. However, it remains unclear whether the same pedagogical strategies and digital tools are equally effective across these domains. For example, visualization dashboards commonly used to display energy consumption or

carbon footprints may not directly translate into social sustainability outcomes, such as equity and inclusion. At the same time, digital platforms could potentially be extended to track and make visible social indicators, such as diversity in project teams or participation patterns in laboratory activities. This raises an important research question for the field: do social sustainability goals require fundamentally different behavioral mechanisms, learning designs, and technological interventions than for environmental sustainability, or can integrated tools support multiple sustainability dimensions simultaneously? Addressing this question is essential for developing comprehensive and balanced sustainability education frameworks and moving towards Society 5.0 and Education 5.0, embedding virtual reality and the metaverse.

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Data Availability

No new data were created or analyzed in this work. Data sharing is not applicable.

Use of Generative AI and AI-Assisted Technologies

AI tools were used for language editing recommendation purposes.

Author Contributions

Yekaterina Kovaleva: Conceptualization, Methodology, Formal analysis, Visualization, Writing – original draft; R. I. Minu: Validation, International viewpoint, Writing – review & editing; Ari Happonen: Conceptualization, Funding acquisition, Supervision, Validation, Writing – review & editing.

Conflicts of Interest

The authors have no conflict of interest to declare.

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